

RESULTS AT A GLANCE

Bigger ≠ Better

Scaling model size often raises energy, latency, and VRAM faster than toy-task quality.

Quantization Changes Economics

q4/q8 variants can cut memory and energy enough to change where mo

OVERVIEW

Everyday LLM calls look simple from Python, but the real deployment decision is a tradeoff between **quality**, **latency**, **VRAM**, **energy**, and **cost**.

We benchmark code generation, summarization, chat, RAG-style search, and embeddings across model sizes, quantization levels, architectures, and CPU/GPU configurations.

Thesis: the most accurate LLM is often not the most economically efficient one. LLM deployment is a systems engineering problem, not just model selection.

BENCHMARK SCOPE

19 models / variants	5 workload families	234 metric rows	L40S NVIDIA L40S
Question	Measured Signals	Why It Matters	
Bigger model?	toy score, latency, joules, VRAM	find diminishing returns	
Quantize?	q4/q8/fp16, tokens/J	fit smaller hardware	
Which workload?	task score by model	avoid one-model thinking	

MEASUREMENT PIPELINE

Each run produces one row of metrics: latency slices, token counts, active-device energy, memory, and workload score.

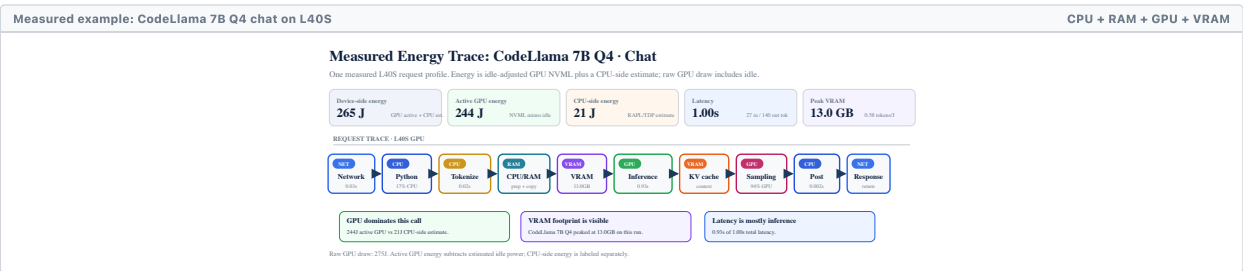
FIXTURES	Repeatable prompts and document sets for each workload.
CLIENT	Ollama/OpenAI-compatible adapter wraps model calls and response logs.
ENERGY	NVML samples GPU power/utilization/VRAM; CPU uses RAPL or labeled TDP estimate.
VISUALS	pandas + matplotlib turn traces into poster-ready figures.

Energy attribution matters: active-device energy is cleaner for model comparison; total wall energy is better for full-machine cost.

WHAT GETS TIMED

Slice	Meaning	Why Python Devs Care
Network	client call overhead	endpoint placement, batching
Tokenization	prompt processing / prefill	long docs are not free
Inference	model generation	model size dominates
Post-processing	parse, serialize, rank	pipeline glue is measurable

MEASURED ENERGY TRACE FOR ONE LLM



This turns the abstract request trace into concrete joules, latency, GPU utilization, and VRAM for one measured LLM call.

WHAT THE NUMBERS MEAN

Toy task score Small reproducible task signal. Example: 0.67 means 2 of 3 deterministic checks passed.	Energy Active-device joules: GPU NVML draw minus idle baseline, or labeled CPU estimate.
Tokens/J Generated or processed tokens divided by active-device energy.	Interpretation Use trends for deployment decisions; this is not a universal model leaderboard.

LEARNING 1: BIGGER MODELS HAVE D

SAME ARCHITECTURE · DIFFERENT SIZES

Scaling parameters often increases infrastru

Same-family comparisons isolate parameter count: Gemma 2B,

Measured benchmark: toy/code score, energy, late

Score note: the quality panel is a tiny code-generat

LEARNING 2: QUANTIZATION CHANG

SAME MODEL · DIFFERENT PRECISION

4-bit and 8-bit models often retain useful qual

Measured benchmark: fp16 vs q8 vs q4 only withi

How to read it: FP16 is the 100% baseline for the ex
only complete same-model FP16/q8/q4 triplets.

Artifacts: benchmark framework, configs, notebooks, CSV metrics, NVML traces, generated PNG/SVG assets, and printable poster.
Open repository: github.com/shivaylamba/pycon-2026-poster | **Inspired by:** Alizadeh et al., *Language Models in Software Development Tasks: An Experimental Analysis of Energy and Accuracy*, arXiv:2412.00329.